

# HILL MODEL

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## 1 Introduction

The phenomenon of muscle contraction can be modeled similarly to a mass-spring-damper system, which is known as the Hill model. Although this was done to simplify simulations and substantiate existing evidence, anatomical and physiological knowledge of the muscle is required. This report provides that background to help understand the graphs of the Hill model presented below.

## 2 Development

Muscle is a striated tissue responsible for the mobility of the skeletal system; its fundamental unit is the myofibril, where the proteins responsible for movement are found. The primary proteins involved in muscle contraction and extension are myosin and actin. In the sarcomere, which are the structural units where myofibrils are arranged, we find the following protein layout: for each thick filament (formed by myosin), there are two thin filaments (formed by actin).

Although thin filaments are composed of other proteins such as troponin and tropomyosin, actin is the one that makes contact with the myosin heads, leading to contraction. The structure of the myofibril and its composition can be seen in the following image: Based on this, we can ask a couple of questions: if the amount of myosin is less than actin, how does that impact movement? This is simple to think about since, during a contraction where the muscle changes its length, it increases in size in a specific area. This is where its name originates, as it means "little mouse" in Greek, because the Romans viewed a contracted muscle as a small mouse moving under the skin.

Therefore, it would be logical to think that for this mass accumulation to occur in one spot, there would be more myosin bound in a place where actin had a higher proportional ratio, such as 2 to 1 or even much higher. Conversely, it would be a 1-to-1 ratio in its relaxed state, and a 2-to-1 or 3-to-1 ratio in its extended state, but now looking at the actin-myosin relationship.

This deduction will later align with the Hill model. Since the relationship between proteins and movement is now somewhat understood, we can address how muscle fibers respond to higher or lower velocity forces and how this relates to a mass-spring-damper model.

## 3 Hill Model (Analogy)

The Hill model was proposed by the 1922 Nobel Prize winner in Physiology, Archibald Vivian Hill. It is a phenomenological model, meaning it resembles a known system from which its analysis

begins, though it is not directly linked to it. Thus, the Hill model conceives muscle contraction as a phenomenon involving a contractile element (CE) and a series elastic element (SEE), along with their respective pennation angle.

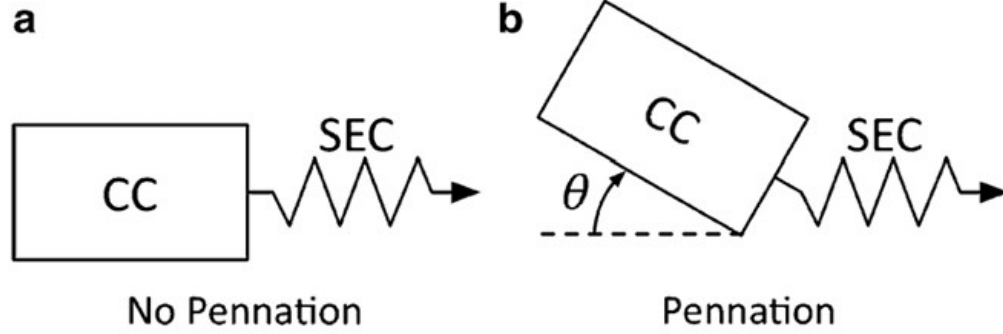


Figure 1: Hill model - fundamental representation

From Hill's observations, the following equations were obtained to describe the force of the elastic element based on the length of the tendon segment under tension ( $L_{SEC}$ ) and when it is not ( $L_u$ ):

$$F_{SEC} = \begin{cases} K_1(L_{SEC} - L_u)^2, & L_{SEC} > L_u \\ 0, & L_{SEC} \leq L_u \end{cases} \quad (1)$$

$$K_1 = \frac{F_o}{(U_o L_u)^2} \quad (2)$$

From this, we can deduce that the tendon generates no force when relaxed, but generates force once it passes that threshold. However, we are interested in analyzing the force of the entire contractile element relative to its length to see how they behave together. Therefore, the following equation proposed by Hatze describes this behavior:

$$F_L = 0.32 + 0.71 \exp \left( -1.112 \left( \frac{L_{CC}}{L_o} - 1 \right) \right) \sin \left( 3.722 \left( \frac{L_{CC}}{L_o} - 0.656 \right) \right) \quad (3)$$

If we plot this equation in MATLAB, we can observe the following behavior:

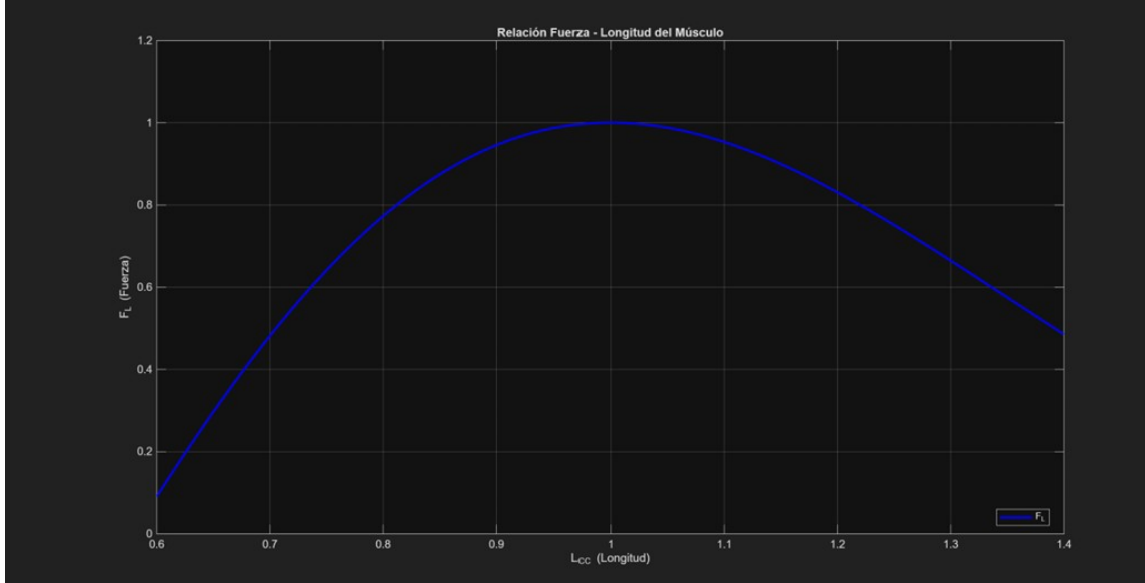


Figure 2: Force vs length in the muscle

From this, we can determine that a contracted muscle produces less force, supporting the hypothesis made at the beginning of this document. It reaches a point where force is greater during isometric contraction, which occurs when its length remains unchanged despite the effort. Finally, its force decreases at full extension because when the muscle exceeds its contractile limit, the tendon takes over to generate that force. The force vs velocity, force vs length, and combined 3D surface graphs will be displayed next to draw the final conclusions.

### 3.1 GRAPH

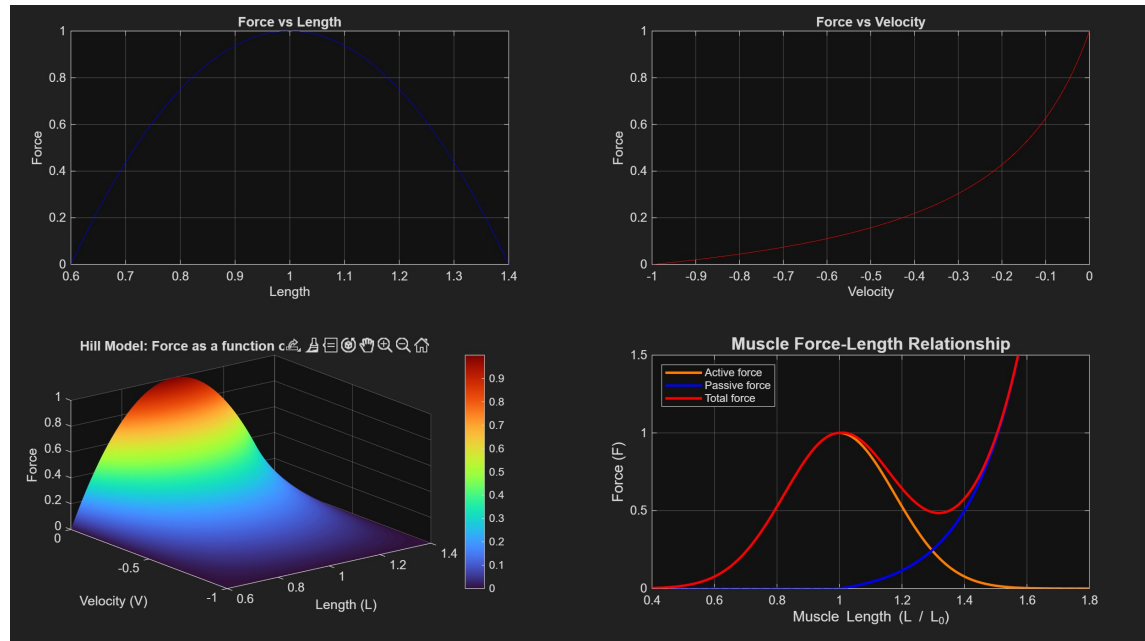


Figure 3: Hill Model Graphics

### 3.2 Conclusions

Based on the previous graphs, we can draw the following conclusions:

- As the muscle contracts or extends, it cannot generate enough force because the myosin and actin cross-bridges maintain a lower ratio. Consequently, their binding points are confined to specific zones, generating concentrated points of force rather than a proportional relationship that would provide a better lever arm for muscle force.
- As the muscle performs faster movements, it progressively loses force, as shown in Graph 2. This happens because higher velocities do not allow enough time for myosin and actin to bind. As a result, the tendons take over by leveraging their elastic properties. This is visible in Graph 4, where as the length of both the muscle and tendon increases, the force shifts from an active component like the muscle (contractile element) to a passive type of force exerted by the tendon.
- Finally, Graph 3 illustrates how length, velocity, and force are distributed across a surface. This integrates these concepts to provide a clearer perspective on how they behave regarding active force.